

Home Work 01: Quotient Rule with Logarithmic Function

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = \frac{x}{\log x}$

নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — TRANSCENDENTAL-ALGEBRAIC QUOTIENT FORMULATION

The problem presents an algebraic linear numerator divided by a natural logarithmic denominator:

$$y = \frac{u(x)}{v(x)}, \quad \text{where } u(x) = x, \quad v(x) = \log x (\equiv \ln x)$$

Applying the fundamental Quotient Rule:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}, \quad \text{with } \frac{d}{dx}(x) = 1, \quad \frac{d}{dx}(\log x) = \frac{1}{x}$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত অন্তরজ সূত্র (Standard Derivative)
লব (u)	x	$\frac{d}{dx}(x) = 1$
হর (v)	log x	$\frac{d}{dx}(\log x) = \frac{1}{x}$
হরের বর্গ (v ²)	(log x) ²	(log x) ² (উল্লেখ্য: (log x) ² ≠ log(x ²))

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

- Logarithmic Base Convention:** In calculus and engineering, unbased log x strictly means natural logarithm ln x (base e).
- Algebraic Cancellation:** Notice that $x \cdot \frac{1}{x} = 1$. A common mistake is omitting 1 and writing only log x in the numerator.
- Exponent Placement:** Denominator is (log x)², not log(x²).

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x-এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{x}{\log x} \right)$$

ধাপ ২: ভাগফলের সূত্র (Quotient Rule) অনুসারে বিস্তার:

$$\frac{dy}{dx} = \frac{\log x \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(\log x)}{(\log x)^2}$$

ধাপ ৩: প্রমিত অন্তরজের মান $\frac{d}{dx}(x) = 1$ এবং $\frac{d}{dx}(\log x) = \frac{1}{x}$ বসিয়ে:

$$\frac{dy}{dx} = \frac{(\log x) \cdot 1 - x \cdot \left(\frac{1}{x}\right)}{(\log x)^2}$$

ধাপ ৪: লবের গুণফল সরলীকরণ করে চূড়ান্ত প্রমিত আকারে রূপান্তর:

$$\frac{dy}{dx} = \frac{\log x - 1}{(\log x)^2}$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = \frac{\log x - 1}{(\log x)^2}$$

 ENGLISH NOTE — DOMAIN, CRITICAL POINTS & SINGULARITY ANALYSIS

- **Domain of $y(x)$:** Logarithm requires $x > 0$, and non-zero denominator requires $\log x \neq 0 \implies x \neq 1$. Thus, $\text{Dom}(y) = (0, 1) \cup (1, \infty)$.
- **Critical / Stationary Point:** Setting $\frac{dy}{dx} = 0 \implies \log x - 1 = 0 \implies x = e$. At $x = e$, the curve has a horizontal tangent where $y(e) = e$. By the second derivative test, this is a local minimum.

 ENGLISH NOTE — GENERAL BASE $\log_a X$ GENERALIZATION

If the problem specified an arbitrary base a (e.g. Common Logarithm base 10):

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a} \implies \frac{dy}{dx} = \frac{\log_a x - \frac{1}{\ln a}}{(\log_a x)^2} = \frac{\ln a \log_a x - 1}{\ln a (\log_a x)^2}$$

In standard AMIE examinations, base e is assumed unless explicitly marked.

 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Quotient Rule Formulation	Correct expansion of $\frac{vu' - uv'}{v^2}$	1.5
Logarithmic Differentiation	Accurate derivative of $\log x \implies \frac{1}{x}$	1.5
Algebraic Simplification	Simplifying $x \cdot \frac{1}{x} = 1$	1.0
Final Standard Format	Writing $\frac{\log x - 1}{(\log x)^2}$	1.0
Total AMIE Section-A Marks		5.0 / 5.0